

Forecasting the Future 2026

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Forecasts

Forecast 1:

There is a 45% chance of professional & Business Services employment falling below its fixed pre-2020 (2010–2019) trend for at least 12 consecutive months at some point before end-2035.

Forecast 2:

There is a 55% chance that, conditional on a significant financial-stress event defined as National Financial Conditions Index (NFCI) greater than 1.0, a Warsh-led Federal Reserve does not deploy balance-sheet intervention within two quarters of onset.

Forecast 3:

There is a 30% chance that, before the end of 2028, the aggregate trailing-4-quarter capex YoY growth across Microsoft, Alphabet, Amazon, and Meta, falls below 10% for at least two consecutive quarters.

Forecast 4

There is a 94% chance that the United States announces at least one new advanced-node semiconductor export-control action targeting sub-5nm logic, advanced fabrication tools, or High Bandwidth Memory technology towards China before the end of 2027.

Forecast 5

There is a 78% chance the US 10-year Treasury Inflation-Protected Securities yield averages above 1.75% over calendar years 2026–2028.

Forecast 6

There is an 80% chance the US data-center electricity demand growth outpaces grid capacity additions through 2028, sustaining elevated wholesale power prices in the major data-center hubs.

Forecast 7

There is a 98% chance the USD/CNY does not break past 7.80 on a sustained basis at any point through end-2030.

Forecast 8

There is a 52% chance that China's non-financial-sector debt/GDP does not exceed 320% beyond two consequent quarters through end-2032.

Forecast 9

There is a 94% chance that China's official foreign exchange reserves remain above \$2,674 billion (no greater than a 20% decline from their high level in the previous three years) through end-2030.

Forecast 10

There is a 16% chance that China achieves volume indigenous production of 5nm-class logic, defined as TSMC-N5 density-equivalent or 100k wafer-starts per month for at least 3 months, by Semiconductor Manufacturing International Corporation /Huawei before the end of 2028.

Forecast 11

There is an 84% chance that China's general-public-budget deficit-to-tax-revenue ratio exceeds 40% in any year 2026–2028.

Forecast 12

There is a 9% chance that China onshore corporate bond defaults exceed 60 defaulting issuers in calendar year 2027.

Holistic Synthesis

The prevailing consensus holds that AI and Mercantilism work in conjunction in a manner that reinforces one another. The race to build artificial intelligence and the drive for national self-sufficiency feed the same engine. This impulse toward independence from allied states is not a modern dilemma, a functioning practice three centuries ago; though it has re-entered the global stage as the priority has changed to owning innovation rather than merely contributing. When we narrow our lens to the development of Artificial Intelligence (AI), we see this phenomenon spotlighted. Its progress is not an isolated national achievement but a production chain distributed across the globe: EUV lithography built in the Netherlands, semiconductors fabricated in Taiwan, data centers spread through Germany, and frontier models spearheaded in the United States and China. That very distribution is what makes mercantilism possible, wherein each stage sits inside a different nation, and each nation now holds a chokepoint it can seize, restrict, or weaponize. What appears to be a coordinated global process is in fact a contest to capture as much of the chain as possible, through export controls, excluded trading partners, and geopolitical leverage. This scramble invited our skepticism. With so much capital and rivalry poured into owning the process, we are compelled to ask whether the innovation will ever amount to a return on its investment.

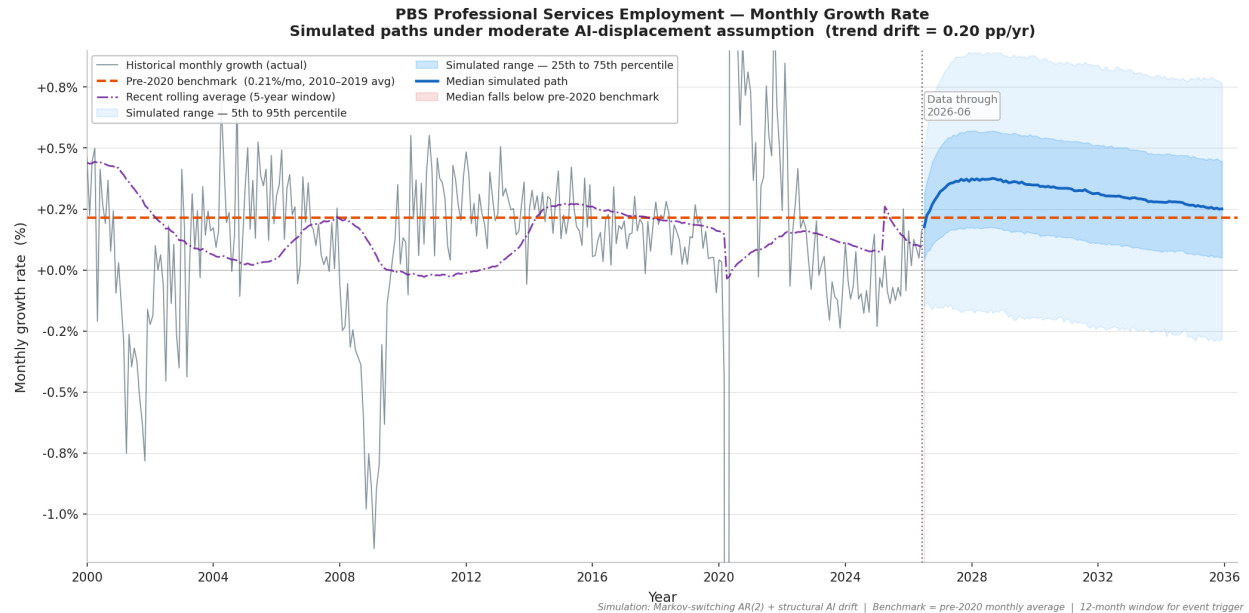
Our thesis is contrarian to this. While much of Wall Street is hopeful of the returns that AI can grant us, we believe the race will fail to justify the capital being poured in: productivity gains will disappoint relative to what markets have priced. More provocatively, we have reason to believe the market has the fragility backwards. It treats the American AI boom as the sturdy, self-financing engine and China's debt-laden build-out as the fragile one. We argue the reverse: China's debt is more absorbable than the crisis consensus assumes, while the United States, increasingly financed on credit and underwritten by a weakening policy backstop, carries the mispriced fragility.

We start with the key global facts in front of us. In recent years, the global savings glut that long held rates down is reversing. Those who were considered historical 'net-savers' states, such as Germany, Japan, and Korea, have begun domestic capital expenditure rather than exporting it into US assets. At the same time, within the United States, AI capital expenditure contributed 1.44pp to GDP growth in 2026 – an astonishing figure contributing to nearly half of overall growth by a singular industry. But the framing is everything. AI is a capital-intensive, labor-light industry, and that combination is more precarious than it appears.

Consider the comparison: investment in manufacturing versus AI. A conventional factory, once built, pays for itself: it produces output, employs workers, and returns the original capital investment. AI's growth has done the first part, consuming staggering amounts of capital. However, it is yet to deliver the second, producing revenue that can work to repay it and is forecasted not to for many years to come. As the timeline progresses, the view is beginning to shift: it is therefore an industry that must be continuously refinanced from the outside rather than one that sustains itself from within. Critically, what happens when the price of that capital rises to meet its scarcity, wherein an industry built on cheap, borrowed money must refinance into an expensive world?

This doubt became the crux of our inquiry. The entire justification for AI investment rests on a single promise, a step-change in productivity, and so productivity is what we set out to forecast. The United States, across the leading hyperscalers, has invested an aggregate of roughly \$700bn. The thesis underwriting those sums is that AI will lift output across every white-collar industry, producing far more work from the same labor. The consensus leans on a comforting sequence: the payoff is real and will arrive, just later, in the shape of a J-curve that dips before it delivers. Rather, we believe the gap bites first.

Capital expenditure is outrunning both realized productivity and the revenue AI actually generates; the coverage between what is being spent and what is being earned deteriorates; and that shortfall surfaces as a financial problem well before it resolves into a productivity win.



Measure of change in professional service employment as a result of AI development

The shortfall starts as a financial one, and its earliest observable trace is in the labor market. For the investment thesis to succeed, we require a simultaneous job displacement and generation of new output, revenue, and new kinds of work to expand the economy on net. However, displacement without a payoff is capital spent to shrink the workforce while producing too little to justify it. That is what we observe. Employment in professional and business services falls below its long-run trend as AI removes roles, but the offsetting new work that past technology cycles produced has not appeared. We are seeing merely the cost without the benefit.

Where the issue rises is that the early growth was largely financed by cheaper debt. So when the returns disappoint, the problem becomes financial. Borrowers who spent big on AI infrastructure still owe that money, and when it comes due, they have to refinance it. To get more money, it becomes increasingly challenging as the cost of capital rises thereafter. If rates are high and returns are thin, that refinancing hurts, and the weakest players start to break.

Normally this is where the Fed steps in. Since 2008, the reflex has been the same: cut rates, expand the balance sheet, cushion the fall. The market assumes it happens again. But that assumption depends on who runs the Fed, Kevin Warsh is unlikely to follow the script. He has spent years attacking exactly this playbook, and it would be out of order for him to cushion the fall. If he lets rates run and lets the stress play out, this makes the strain worse.

Crucially, the problem described is by no means an American problem. In fact, the consensus believes that the impacts faced by China would be substantially greater than the American dilemma. Instead, we believe China will absorb the stress rather than spiral into a debt crisis, unlike the consensus.

To see why, consider Japan's 1980s bubble as a case study. Its stark growth during the technology manufacturing boom was funded by high household saving levels of ~15%. So when it burst, the debt was owed by Japanese citizens to Japanese institutions. It did not flee; the yen did not crash, and the state had room to absorb the shock slowly. Japan got stagnation, not collapse. China rhymes on what matters: gross savings near 43% of GDP, household deposits of roughly ¥165tn, and debt that is overwhelmingly domestic and behind capital controls.

We do not overclaim, though, as we want to highlight China's structural issues: a shrinking workforce, a property crash running alongside the build-out rather than before it, and lower incomes per capita causing a thinner safety net. However, functionally, its funding sources are owed inward and cannot run for the exit, which makes it the absorbable kind of debt, not the crisis-prone kind owed to foreign creditors who can pull out overnight.

This durability reshapes the conflict. A straining United States, facing a China that refuses to crack, does not resolve its own problem. It escalates, tightening the screws on the one axis where it holds leverage: chips. Export controls on advanced nodes are near-certain to widen. But the response deepens the loop rather than closing it, because each new restriction pushes China harder toward self-sufficiency, into domestic substitution through SMIC and Huawei. If that substitution succeeds even partially, it proves the very durability we describe and invites still tighter controls. Underneath it all sits the same physical collision from the start: US defense and AI spending and Chinese capex bidding for the same power, the same chips, the same capital. The mercantilist ratchet means neither side can stop.

By the end of 2028, then, the picture is not the clean one the market expects, where the US holds firm and China breaks. The honest resolution is split. The United States carries the more acute fragility, in credit stress, rate pressure, and political rupture, while China carries the more chronic drag, in demographics, deflation, and the slow grind of stagnation. But the debt blowup priced into China does not arrive, and the productivity payoff priced into the US arrives late, if at all. Both consensus bets are wrong. The fragility was never where the market was looking. In its strongest form, the picture tilts further still: if China's capex clearly pays for itself while US returns clearly disappoint, the durability gap widens and US escalation only sharpens. For an investor, the conclusion follows plainly. It argues for stepping back from the crowded US-AI-equity trade, against the China-crisis short, and toward real assets and geographic spread. The lesson is not that one side wins. It is that resilience, not concentration, is what survives a world where the fragility sits exactly where no one is looking.

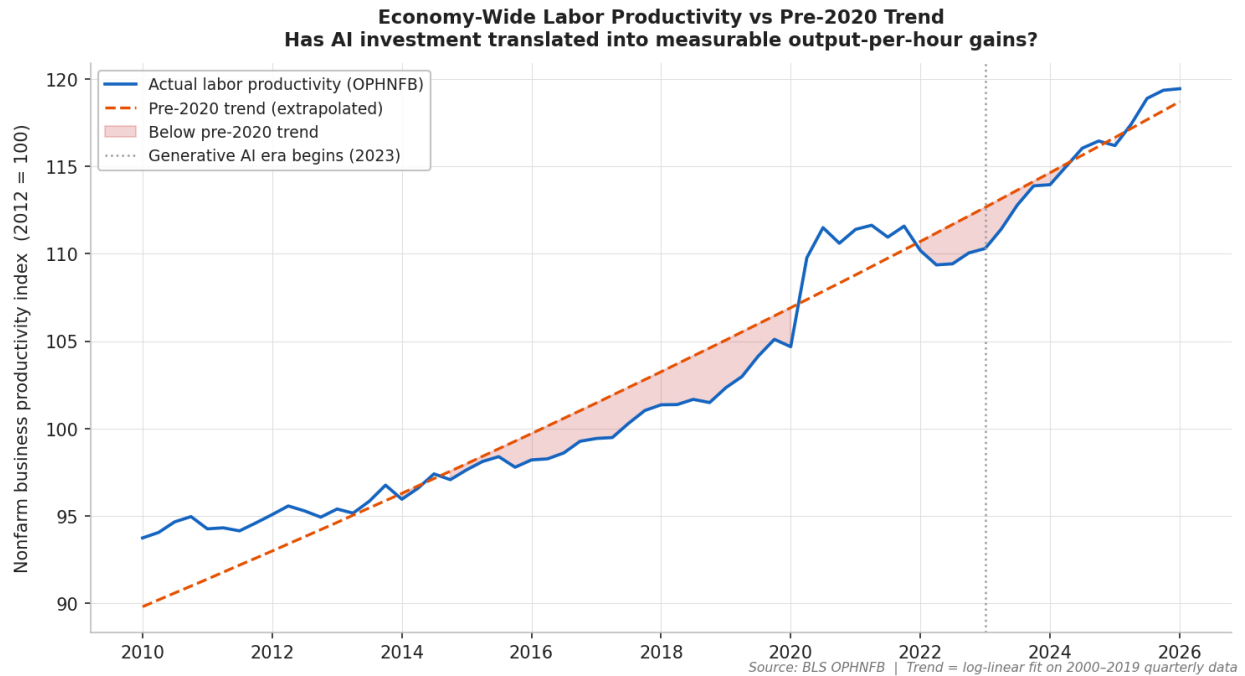
Analytical Appendix

The majority of the forecasts formed used models that are analytical models that assess the continuous change of the current situation by adjusting drift variables to accurately simulate events that follow. However, **Forecast 5** and **Forecast 6** were structural calculations that were merely reasoned rather than simulated, which we want to explain first.

Forecast 5 and Forecast 6 were not continuous measures, but rather, observable structural variables that do not involve a data-generating process. Therefore, we avoided a Monte Carlo process as we believe it would have led to a false prediction. Instead, we chose an appropriate base rate and then adjusted for a structural shift that could be roughly quantified from anecdotal evidence. On **Forecast 5**, the base case is the regime history: the 10-year real yield averaged roughly 2% before 2008, sat near zero through the 2010–2021 savings-glut era, and stands at 2.37% today. The savings glut is now reversing as former net-savers spend at home rather than recycling surpluses into Treasuries, which pushes the elevated regime to persist. As we started above our 1.75% threshold, and with only a returning glut or a hard recession able to pull it back below, we landed at a range of 75%-80%, and we chose to take the median of such. On **Forecast 6**, we followed a similar logic chain: US data-center demand is set to more than double by 2027 while total generation capacity grows low-single-digits. Therefore, the shortfall concentrates in exactly the hubs where AI load is heaviest. Then, as grid buildout is permitting- and interconnection-constrained and cannot accelerate to close the gap by 2028, we put the demand-supply imbalance persisting at roughly 80%.

While we do admit this may be a slightly naïve approach to the prediction, we do believe that this would be a rough approximation of the real probability. Though, much of the other forecasts used existing tools and model construction to build our reasoned probability from the ground up. Consider **Forecast 1** wherein we attempt to understand the displacement caused by AI development. this, we utilize a two-regime Markov Switching Model (akin to the Regime Switching Model Hamilton Filter (1989)) to divide between a calm and an expansion regime, allowing the data to decide which state the economy is in at any point based on the current state by developing a Markov Process. This pattern shift allows us to understand how employment levels would change as we expand the timeline to 2035. Then, we accounted for the structural drift term from AI growth, which allows us to model how the expansion regime's trend gradually declines as AI absorbs white-collar tasks over time. This drift was sourced from the occupational exposure estimates in Eloundou et al. (2023), and weighted across mild, moderate, and aggressive scenarios as advised. With this accounted for we devised a Markov Switching Model that would allow us to simulate employment levels till 2035. Then, we would flag any path where employment would stay below its 2010-2019 trend for twelve consequent months. This would remove any issues of random noise in the event of miscellaneous events causes a short-term employment change and isolating the triggers we wanted to measure.

When working with the model we got an interested result. When setting the drift to zero and the model returns a 29% probability. This figure reflects pure disaster luck, capturing only how often the contraction regime fires by chance. This did not appropriately account for what we wanted to measure. Therefore, we then apply the moderate drift, and the probability rises to 45%. This 16pp gap is the AI-displacement channel on its own, separated from the ordinary risk of a recession. We wanted to highlight the separation as more interesting than the final probability the increment is the AI signal alone being a substantial part of the prediction probability.



U.S. nonfarm business labor productivity relative to pre-2020 trend, 2010–2025

However, we wanted to mention one caveat. The drift is a scenario assumption drawn from exposure research, not a value fit from labor data, because no AI-displacement episode yet exists in the series to estimate from. For this reason, we report the full scenario range rather than a single point, and we treat the aggressive case as an upper bound rather than our central view.

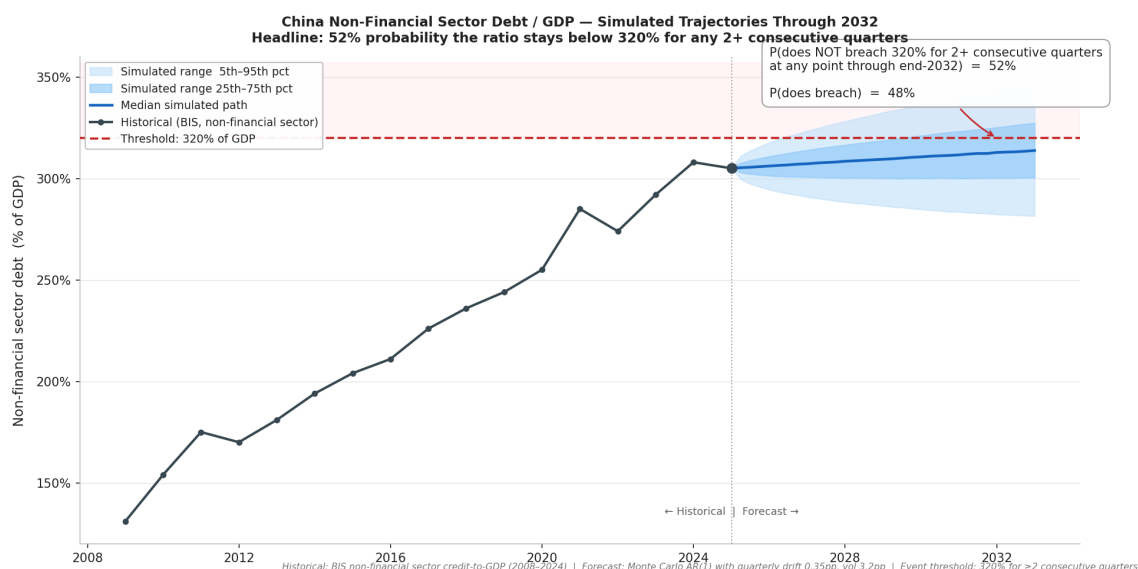
Secondly, consider **Forecast 2** where we attempt to forecast the Federal Reserve’s (Fed) intervention and how Kevin Warsh’s more hawkish strategy would affect the AI CapEx growth. For this, we used a conditional reaction-function model as it was a clear situation of having a condition met and a necessary reaction that would/should take place. Crucially, we wanted to understand what the difference would be between previous Fed Chairmen and Kevin Warsh’s strategy would be. Firstly, we established the condition of a crisis as we wanted to be as binary as possible. This would avoid the question of whether there would be a crisis in the first place. Then, to understand the response, we split into two independent tools: rate cuts versus balance-sheet intervention. The 2023 Silicon Valley Bank episode proves these are separate; the Fed stood up an emergency facility in 72 hours while still hiking rates. This separation means the tools are independent levers, not coupled responses.

Then, we established what a typical chair would do across every modern stress episode, drawing from 2008, 2020, and 2023. A typical chair deploys balance-sheet intervention in roughly 90% of crises. Warsh, calibrated from his own record as a governor, sits at approximately 45%, with a range of 35 to 55%. This roughly 45pp gap is measured as the difference in hawkish nature that he has had in prior records and expanded as a measure towards his control as the chairman. The significance here is not simply that Warsh is a hawk, because the thesis does not need just any hawkish chair. Instead, we want to highlight the specific mechanism. We believe he would cut rates yet withhold the backstop that contains modern crises, which is exactly the stress-amplifying configuration the thesis requires. We do acknowledge this carries real uncertainty. The 45%

is our single biggest judgment call. It rests on a tension in Warsh's record. He stated publicly that he opposed quantitative easing and that the balance sheet was bloated. These words suggest he would refuse to deploy these tools as chair. Yet his record as governor points the other way. He voted for every crisis measure that came before him. This behavior suggests he would act differently when actual stress arrives. We weigh his stated positions more heavily because, as chair, he cannot defer to a group consensus. But his revealed behavior makes the opposite case plausible, and a reader could reasonably weight it differently.

Thirdly, we want to expand on **Forecast 8**, wherein we wanted to understand the leverage identity of China. This process was to understand the increase in debt growth by China when increasing its expenditure on AI growth. This was quantifiable due to China's economic structure. For this, we test the second derivative of the scale, measuring the rate of increase of debt to understand whether the change is steady or if it is accelerating debt. If we were to test whether the debt would stay below a certain threshold as debt grows alongside, and at a faster rate than GDP, and therefore would be resolved by default. A key comparison we wanted to make was to understand whether the leveraging cycle would function similarly to the Japan case in the early 2000s during the rise of the technology manufacturing industry. In this case example, the leveraging was roughly steady and housed predominantly in household debt. Hence, the mechanism we employed was simulating the annual percentage-point change in leverage forward across productive, mildly unproductive, and dead-weight ICOR scenarios, and asking whether that annual increment stays steady or trends upward in a sustained way. Our acceleration analysis is what identifies 320% as the threshold beyond which the increase would no longer be absorbable.

However, we got genuine uncertainty in our result. We got an approximate 51% as our probability, narrowing down the sustainable self-financing chance to a coin flip. This distinction is what lets us claim China is durable but stressed rather than simply fine. This framing is more defensible because it concedes the real strain in the system while still denying the sudden collapse the market prices. We do add one caveat here. The result turns on whether ICOR, the incremental capital-output ratio, rises over time in a dead-weight spiral or holds steady on the Japan-path. This is the same kind of judgment call as the EUV question in our chip forecasts, the single hinge on which the outcome swings. We lead with the slope-only number that produces the 51%, because the stricter joint test would penalize ordinary volatility rather than genuine acceleration.



Simulated trajectories of China's non-financial sector debt as a share of GDP against the 320% threshold through 2032

Now, we can focus on the shorter supplements. These are crucial in building up the larger elements of the forecast bundle. Firstly, we regard the China durability cluster. These rest on four independent tests, each asking whether an acute-crisis event fires. **Forecast 7** resolves on whether daily USD/CNY closes above 7.80 on a sustained basis through 2030, and we price a break at just 2% because the debt is domestic and trapped behind capital controls; therefore, it is improbable that it flees the way foreign-owned debt does. This is vital in underpinning the securing of CapEx available by the state and, in consequence, AI growth. **Forecast 9** resolves on whether reserves fall 20% below their trailing three-year peak, and we price a drawdown at roughly 6% using a two-regime model in which the large buffer and short historical crisis duration keep most paths intact. **Forecast 12** resolves on whether distinct onshore corporate defaults exceed sixty issuers in 2027 on a single fixed data source, and our negative-binomial count across property scenarios places this at 9%, with measurement risk our main caveat. **Forecast 11** resolves on whether the official deficit-to-tax ratio breaches 40% in any year, and we price this at 84%, noting that the augmented measure including land revenue sits near 100%. The value here is the cross-consistency, as four separate tests point to the same conclusion: acute crisis is unlikely, but the stress is real and routed off the headline figures.

The chip forecasts form a matched pair, and the interaction between them is the point. **Forecast 4** resolves on whether the US Bureau of Industry and Security publishes at least one new advanced-node control before 2027, and a Poisson model on a base rate of fifteen qualifying actions since 2018 places this at 94%. **Forecast 10** resolves on whether China reaches volume indigenous 5nm-class production by 2028, judged on physical teardown and shipment evidence, and we price this at only 16% because the yield wall on older lithography is a hard physical constraint. These two numbers drive the escalation loop, as controls tighten precisely because containment is uncertain, and each Chinese advance invites a further restriction rather than a resolution.

Finally, **Forecast 3** resolves on whether aggregate trailing-four-quarter hyperscale capex growth falls below 10% for two consecutive quarters before 2028, and we price this at 30% through a trend-break method, a modest number because the boom is still accelerating and the spending-versus-returns gap is therefore widening rather than closing.